Complex Geometry--Huybrechts

Local theory

Complex manifolds

Kahler Manifolds

- Kahler identity
- Hodge theory on Kahler manifolds
- Lefschetz theorem

Vector Bundles

- Hermitian vector bundles and Serre duality
- connection
- curvature
- Chern class

Applications of cohomology

- Hizebruch-Riemann-Roch theorem
- Kodaira vanishing theorem
- Kodaira embedding theorem

Deformation of complex structures

- Maurer-Cartan equation
- general results

Algebraic Geometry--Hartshorne

Variety

Schemes

- sheaves of modules
- divisors, projective morphisms
- differentials
- formal schemes

Cohomology

- cohomology of sheaves
- Serre duality
- higher direct images of sheaves
- flat/smooth morphisms

Curves

- Riemann-Roch theorem, Hurwitz's theorem
- embeddings in projective space

Surfaces

- pairing on surface etc.

Abelian Varieties--Lange

- Complex Tori
- Line bundles(factos of automorphy, first Chern class of a line bundle)
- Appell-Humbert theorem and canonical factors
- The dual complex torus and the Poincare bundle

- h^0(L) for a positive line bundle L(theta function, semi-positive case)
- Cohomology of line bundles(harmonic forms with values in L, vanishing theorem)
- Riemann-Roch theorem(analytic, geometric)

Algebraic Number Theory--lecture notes by Yiwen Ding

- algebraic number fields, trace & norm, class number, ramification
- local fields, absolute value, different and discriminant, place
- adeles & ideles
- zeta functions, Dirichlet L-functions

Class Field Theory--lecture notes by Yiwen Ding

- Lubin-Tate formal group
- group cohomology
- local class field theory, Brauer group
- global class field theory, global reciprocity law

Lie Groups and Lie Algebras-Spensaki

- compact Lie groups
- basic representation theory
- harmonic analysis, Peter-Weyl theorem
- Lie algebras, exponential map
- maximal tori and cartan subalgebras, structure of compact Lie group
- root theory, killing form, lattice, Weyl group, ABCDEFG
- highest weight theory, Weyl character formula, Borel-Weil theorem

J.P. Serre--A course in Arithmetic

Algebraic methods

- finite field, quadratic reciprocity law
- hilbert symbol
- quadratic forms over Q_p and over Q

Modular forms

- modular group, modular function, space of modular forms
- Hecke operator, theta functions

Intersection Theory--William Fulton

- multiplicity, normal cones, Bezout's theorem, intersection products
- divisors, rational equivalence
- Chern class & Segre class
- Gysin maps, Chow ring, Grassmannians and flag varieties

Deformation Theory--Hartshorne

First order deformation

- T^i functor
- infinitesimal lifting property
- deformation of rings

Higher order deformation

- subschemes
- vector bundles
- schemes

- obstruction theory in above cases

Formal Methods

- functors of Arin rings, miniversal/universal
- Schlessinger's criterion, pro-representability
- deformation functor of schemes/line bundles

Homology Theory

- singular (co)homology, Mayer-Vietoris pair
- relative (co)homology, celluar homology
- Euler characteristic
- universal coefficient theorem
- products of complexes, cup/cap product, Kunneth formula
- Poincare duality, Lefschetz duality, Alexander duality